QM Basic

Formula Sheet - Lectures 1–10

Notation used throughout

Ω	sample space
A, B	events
A^c	complement of A
X, Y	random variables
x_i, y_i	observed values in row i
n, N	sample size, population size
μ, σ^2, p	population mean, variance, proportion
$\bar{X}, S^2, \widehat{P}$	estimators
$\bar{x}, s^2, \hat{p}$	realised estimates
α	significance level
df	degrees of freedom

$$H_0 : \theta = \theta_0 \qquad \text{versus} \qquad H_1 : \theta \neq \theta_0 \quad \text{(two-sided)}$$

Lecture 1 - Data summaries and units

$$\text{proportion} = \frac{n_A}{n}, \qquad \text{percentage} = 100\frac{n_A}{n}\%.$$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i, \qquad \text{range} = \max_i x_i - \min_i x_i.$$

$$\text{percentage-point change} = b - a,$$

$$\text{relative percentage change} = 100\frac{b-a}{a}\% \quad (a \neq 0).$$

Lecture 2 - Probability foundations

$$0 \leq P(A) \leq 1, \qquad P(\Omega) = 1, \qquad P(A^c) = 1 - P(A).$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

$$A \cap B = \varnothing \quad \Longrightarrow \quad P(A \cup B) = P(A) + P(B).$$

$$P(A) = \frac{|A|}{|\Omega|} \quad \text{(equally likely outcomes)}.$$

$$X : \Omega \to \mathbb{R}.$$

$$B_i \cap B_j = \varnothing \ (i \neq j), \qquad \bigcup_{i=1}^{k} B_i = \Omega.$$

Lecture 3 - PMFs, CDFs, and expectation

$$p_X(x) = P(X = x), \qquad p_X(x) \geq 0, \qquad \sum_{x \in \mathcal{X}} p_X(x) = 1.$$

$$F_X(a) = P(X \leq a) = \sum_{x \leq a} p_X(x).$$

$$P(a < X \leq b) = F_X(b) - F_X(a).$$

$$E[X] = \sum_{x \in \mathcal{X}} x\, p_X(x), \qquad E[g(X)] = \sum_{x \in \mathcal{X}} g(x) p_X(x).$$

$$X \sim \text{Bernoulli}(p) : \qquad P(X = 1) = p, \quad P(X = 0) = 1-p, \quad E[X] = p.$$

Lecture 4 - Variance and conditioning

$$\mu = E[X], \qquad \text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \{E[X]\}^2.$$

$$\text{SD}(X) = \sqrt{\text{Var}(X)}.$$

$$X \sim \text{Bernoulli}(p) : \qquad \text{Var}(X) = p(1-p).$$

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad s = \sqrt{s^2}.$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(A \cap B) = P(A \mid B)P(B), \qquad P(B) > 0.$$

$$P(A) = \sum_{i=1}^{k} P(A \mid B_i)P(B_i).$$

$$P(B_i \mid A) = \frac{P(A \mid B_i)P(B_i)}{\sum_{j=1}^{k} P(A \mid B_j)P(B_j)}.$$

Lecture 5 - Joint PMFs and correlation

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

$$p_X(x) = \sum_{y} p_{X,Y}(x, y), \qquad p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}.$$

$$X \perp Y \quad \Longleftrightarrow \quad p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \text{for every } x, y.$$

$$A \perp B \quad \Longleftrightarrow \quad P(A \cap B) = P(A)P(B).$$

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - E[X]E[Y].$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\text{SD}(X)\,\text{SD}(Y)}, \qquad -1 \leq \rho_{XY} \leq 1.$$

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}), \qquad r_{XY} = \frac{s_{XY}}{s_X s_Y}.$$

Lecture 6 - Continuous distributions and quantiles

$$f_X(x) \geq 0, \qquad P(X = x) = 0, \qquad P(a < X \leq b) = F_X(b) - F_X(a).$$

$$X \sim U(a,b): \qquad f_X(x) = \begin{cases} \dfrac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise}, \end{cases}$$

$$P(c < X < d) = \frac{d-c}{b-a}, \qquad E[X] = \frac{a+b}{2}, \qquad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

$$Q_1 = q_{0.25}, \qquad Q_2 = q_{0.50}, \qquad Q_3 = q_{0.75}, \qquad \text{IQR} = Q_3 - Q_1.$$

$$X \sim N(\mu, \sigma^2), \qquad Z = \frac{X - \mu}{\sigma} \sim N(0,1).$$

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68, \qquad P(\mu - 1.96\sigma \leq X \leq \mu + 1.96\sigma) \approx 0.95.$$

Lecture 7 - Samples and estimators

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \widehat{P} = \frac{1}{n}\sum_{i=1}^{n} X_i \quad (X_i \in \{0,1\}).$$

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \sum_{i=1}^{n}(X_i - \bar{X}) = 0, \qquad df = n-1.$$

$$n \leq 0.10N \quad \text{(sampling without replacement)}.$$

Lecture 8 - Sampling distributions and SEs

$$E[\bar{X}] = \mu, \qquad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} \approx \frac{S}{\sqrt{n}}.$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \mathrel{\dot\sim} N(0,1), \qquad \bar{X} \mathrel{\dot\sim} N\left(\mu, \frac{\sigma^2}{n}\right).$$

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

$$E[\widehat{P}] = p, \qquad \text{SE}(\widehat{P}) = \sqrt{\frac{p(1-p)}{n}}.$$

$$np \geq 10, \qquad n(1-p) \geq 10.$$

Lecture 9 - Confidence intervals and zero tests

$$\text{estimate} \ \pm \ \text{critical value} \times \text{SE}.$$

$$\bar{X} \pm t_{1-\alpha/2, n-1}\frac{S}{\sqrt{n}}.$$

$z_{0.975} = 1.960$,

df	5	10	20	30	60	∞
$t_{0.975,df}$	2.571	2.228	2.086	2.042	2.000	1.960

$$\delta = \mu - \mu_0, \qquad \widehat{\delta} = \bar{X} - \mu_0, \qquad T = \frac{\widehat{\delta}}{S/\sqrt{n}}, \qquad df = n-1.$$

$$H_0: \delta = 0 \qquad \text{versus} \qquad H_1: \delta \neq 0.$$

$$p = P(|T_{df}| \geq |t_{obs}| \mid H_0), \qquad p \leq \alpha \implies \text{reject } H_0.$$

Lecture 10 - Tests, errors, and power

$$H_0: p = p_0 \qquad \text{versus} \qquad H_1: p \neq p_0.$$

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}, \qquad p\text{-value} = 2P(Z \geq |z_{obs}|).$$

$$np_0 \geq 10, \qquad n(1-p_0) \geq 10.$$

$$\hat{p} \pm z_{1-\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

$$P(\text{Type I error}) = \alpha, \qquad \text{Power} = P(\text{reject } H_0 \mid H_1 \text{ is true}).$$

$$t = \frac{b_j}{\text{SE}(b_j)}, \qquad b_j \pm t_{1-\alpha/2, df}\,\text{SE}(b_j).$$

$$H_0: \beta_j = 0 \qquad \text{versus} \qquad H_1: \beta_j \neq 0.$$